



Scenario-Based Interview Questions

For Senior DevOps / GCP / Platform / MLOps / AI Infra Engineers

These are MUCH closer to real Google / senior-level interviews than theoretical questions.

Senior interviews often sound like:

“Production is failing. What would you do?”

not:

“What is Kubernetes?”



KUBERNETES — SCENARIO QUESTIONS

Scenario 1 — Pods Randomly Restarting

A production application on GKE starts restarting randomly every few hours.

Users report intermittent failures.

CPU looks normal.

Discuss

- investigation steps
- memory analysis
- OOMKilled
- probes
- node pressure
- logs/events
- container limits
- debugging commands

Scenario 2 — Ingress Suddenly Stops Routing

Traffic reaches load balancer but requests fail intermittently.

Explain

- ingress debugging
 - controller logs
 - service endpoints
 - DNS
 - TLS issues
 - NetworkPolicy
 - readiness probes
-

Scenario 3 — Cluster Scaling Failure

HPA is configured but pods are not scaling during traffic spikes.

Investigate

- metrics server
 - resource requests
 - HPA events
 - Cluster Autoscaler
 - pending pods
 - node quotas
-

Scenario 4 — Multi-Tenant Cluster Security

Multiple teams share one Kubernetes cluster.

One team accidentally accessed another team's resources.

Design

- RBAC
- namespaces
- network policies

- admission policies
 - resource quotas
 - workload identity
-

Scenario 5 — Node Becomes NotReady

One node in production becomes NotReady during peak traffic.

Explain

- pod eviction
 - node diagnostics
 - kubelet logs
 - autoscaler behavior
 - workload redistribution
 - HA strategies
-



GCP + GKE — SCENARIO QUESTIONS

Scenario 1 — Massive GCP Bill Spike

Your monthly GCP cost suddenly increases by 40%.

Investigate

- billing analysis
 - logging costs
 - egress traffic
 - idle resources
 - oversized nodes
 - GPU usage
 - autoscaling inefficiencies
-

Scenario 2 — GKE Cluster in Private Network Cannot Pull Images

Debug

- Cloud NAT
 - Artifact Registry permissions
 - DNS
 - firewall rules
 - service account scopes
-

Scenario 3 — Multi-Region Failover

One GCP region fails completely.

Explain

- traffic failover
 - global load balancing
 - database replication
 - DNS failover
 - disaster recovery
 - RPO/RTO
-

Scenario 4 — IAM Misconfiguration

A service account accidentally got Owner access in production.

Explain

- risk assessment
 - remediation
 - audit logs
 - least privilege redesign
 - org policy controls
-

Scenario 5 — GKE Upgrade Breaks Production

After cluster upgrade, workloads fail unexpectedly.

Investigate

- deprecated APIs
 - node compatibility
 - ingress changes
 - CNI issues
 - rollback strategies
-

TERRAFORM — SCENARIO QUESTIONS

Scenario 1 — Terraform State Corruption

Terraform state file becomes corrupted during deployment.

Explain

- recovery strategy
 - backups
 - remote state
 - locking
 - importing resources
-

Scenario 2 — Multiple Teams Modify Infra Simultaneously

Design

- state isolation
- CI/CD workflows
- policy checks
- environment separation

- module strategy
-

Scenario 3 — Drift Detected in Production

Someone manually changed infrastructure in GCP console.

Discuss

- drift detection
 - remediation
 - governance
 - prevention strategies
-

Scenario 4 — Terraform Deployment Accidentally Deletes Production Resource

Explain

- blast radius reduction
 - lifecycle rules
 - approvals
 - recovery
 - rollback design
-



PYTHON + FASTAPI — SCENARIO QUESTIONS

Scenario 1 — FastAPI Service Becomes Slow Under Load

Investigate

- async issues

- blocking calls
 - DB bottlenecks
 - thread exhaustion
 - profiling
 - caching
-

Scenario 2 — API Memory Usage Keeps Growing

Explain

- memory leak debugging
 - connection pools
 - object retention
 - monitoring tools
-

Scenario 3 — Thousands of Concurrent Requests

Design

- async architecture
 - rate limiting
 - queues
 - autoscaling
 - caching
 - load balancing
-

Scenario 4 — API Needs Zero Downtime Deployment

Explain

- rolling deployments
 - canary
 - blue-green
 - backward compatibility
 - schema migrations
-



MLOPS — SCENARIO QUESTIONS

Scenario 1 — Model Accuracy Suddenly Drops

Business metrics fall significantly after deployment.

Investigate

- model drift
 - data drift
 - feature drift
 - training-serving skew
 - observability
-

Scenario 2 — GPU Costs Become Too High

Optimize

- autoscaling
 - batching
 - quantization
 - spot instances
 - inference optimization
-

Scenario 3 — ML Pipeline Randomly Fails

Debug

- orchestration
 - dependencies
 - artifact storage
 - retries
 - reproducibility
-

Scenario 4 — Real-Time Inference Latency Spikes

Explain

- autoscaling
 - batching
 - queueing
 - caching embeddings
 - cold starts
-

Scenario 5 — Multiple Teams Deploy Models Simultaneously

Design

- model registry
 - versioning
 - rollout strategy
 - approval workflow
 - monitoring
-

LLMOPS + AI AGENTS — SCENARIO QUESTIONS

Scenario 1 — RAG System Gives Hallucinated Answers

Investigate

- retrieval quality
- chunking
- embeddings
- reranking
- prompt engineering
- context limits

Scenario 2 — AI Agent Loops Forever

Debug

- recursion limits
- tool failures
- memory corruption
- orchestration logic
- timeout handling

Scenario 3 — Vector Database Latency Increases

Explain

- indexing
- shard design
- embedding size
- caching
- retrieval optimization

Scenario 4 — Sensitive Data Appears in LLM Responses

Discuss

- PII masking
- prompt injection
- RBAC
- retrieval filtering
- audit logging

Scenario 5 — LLM API Costs Explode

Optimize

- caching
 - smaller models
 - prompt compression
 - batching
 - hybrid retrieval
-



SYSTEM DESIGN — SCENARIO QUESTIONS

Scenario 1 — Design Global AI Inference Platform

Include

- GKE
 - autoscaling
 - GPU scheduling
 - load balancing
 - observability
 - failover
-

Scenario 2 — Design Multi-Region Kubernetes Platform

Explain

- traffic routing
 - cluster federation
 - DR strategy
 - consistency
 - networking
-

Scenario 3 — Observability Platform Design

Design

- logs
 - metrics
 - tracing
 - alerting
 - storage scaling
-

Scenario 4 — CI/CD Platform for Thousands of Engineers

Discuss

- scaling
 - isolation
 - security
 - caching
 - runner architecture
-

Scenario 5 — Large-Scale RAG Platform

Include

- ingestion
 - vector DB
 - retrieval
 - caching
 - observability
 - cost optimization
-

BEHAVIORAL SCENARIOS (VERY IMPORTANT)

Incident Scenario

Production is down.

Leadership is asking for updates every 10 minutes.

Engineers are panicking.

Explain

- communication
 - prioritization
 - mitigation
 - stakeholder handling
 - postmortem process
-

Conflict Scenario

Senior developer refuses operational best practices.

Discuss

- conflict resolution
 - influence
 - collaboration
 - escalation balance
-

Ownership Scenario

Critical migration has no clear owner.

Explain

- leadership
 - planning
 - execution
 - risk management
-



HOW TO PRACTICE THESE

Record Answers Like This

Structure:

1. Clarify Problem
2. Identify Impact
3. Gather Data
4. Form Hypotheses
5. Debug Systematically
6. Mitigate Fast
7. Root Cause Analysis
8. Prevention Strategy
9. Monitoring Improvements
10. Long-Term Architecture Fix

This sounds VERY senior-level.



MOST IMPORTANT THING

Senior interviews are NOT testing:

✗ memorization

They test:

- ✓ reasoning
- ✓ debugging
- ✓ prioritization
- ✓ communication

- ✓ tradeoffs
- ✓ architecture maturity
- ✓ calm thinking under ambiguity

That's why scenario preparation is extremely valuable for your experience level.